Lab no. 7

Docker Setup and PostgreSQL Containers

NAME: SANIA SHAFIQUE

REGISTRATION NO.: 2023-BCS-078

COURSE CODE: CS-649

SECTION: A

GITHUB USERNAME: Sania270804

Task 1: Install Docker and Docker Compose

```
sania270804@ubuntu:~$ docker --version
Docker version 29.1.3, build 29.1.3-0ubuntu4.1
sania270804@ubuntu:~$ docker compose version
Docker Compose version v5.6.1
sania270804@ubuntu:~$
```

Install Docker and Docker Compose.

Task 2: Run Two PostgreSQL Versions in Docker

```
sania270804@ubuntu:~$ docker pull postgres:9.6
9.6: Pulling from library/postgres
4b9a74ac6ea0: Pull complete
2d87c3a4038c: Pull complete
4e2c4902efbd: Pull complete
01c676df543a: Pull complete
1cb79db8a9e7: Pull complete
f955a6cf127b: Pull complete
1e3d335ef0b7: Pull complete
f62dc55c568d: Pull complete
e8622b8cb6f3: Pull complete
11087f2b0d87: Pull complete
d6d74bba3a57: Pull complete
f6bae7873dd7: Pull complete
8f7722dc50a7: Pull complete
874d4d2a09fd: Pull complete
Digest: sha256:caddd35b05cdd56c614ab1f674e63be778e0abdf54e71a7507ff3e28d4902698
Status: Downloaded newer image for postgres:9.6
docker.io/library/postgres:9.6
sania270804@ubuntu:~$ docker run  postgres:9.6
Error: Database is uninitialized and superuser password is not specified.
       You must specify POSTGRES_PASSWORD to a non-empty value for the
       superuser. For example, "-e POSTGRES_PASSWORD=password" on "docker run".

       You may also use "POSTGRES_HOST_AUTH_METHOD=trust" to allow all
       connections without a password. This is *not* recommended.

       See PostgreSQL documentation about "trust":
       https://www.postgresql.org/docs/current/auth-trust.html
sania270804@ubuntu:~$
```

Pull the PostgreSQL 9.6 image. Try to run the image without setting a password

```
sania270804@ubuntu:~$ docker run -d -e POSTGRES_PASSWORD=password postgres:9.6
7e29138585bdcd381f9d8f59eee9976965ad95d878de0a502c12dd718cd027cb
sania270804@ubuntu:~$ docker ps
CONTAINER ID   IMAGE                    COMMAND                  CREATED         STATUS
    PORTS                                                                                NA
MES
7e29138585bd   postgres:9.6             "docker-entrypoint.s…"   5 seconds ago   Up 4 seconds
    5432/tcp                                                                             st
range_leakey
651a396673d5   0b3563949d36             "/init"                  5 days ago      Up About a minut
e   3000/tcp, 0.0.0.0:3001->3001/tcp, [::]:3001->3001/tcp                               in
tellij-idea-docker-intellij-1
b3bcf179ef56   sonatype/nexus3:latest   "/opt/sonatype/nexus…"   5 days ago      Up 54 seconds
    0.0.0.0:8007->8007/tcp, [::]:8007->8007/tcp, 0.0.0.0:8006->8081/tcp, [::]:8006->8081/tcp   ne
xus
sania270804@ubuntu:~$
```

Start a new PostgreSQL 9.6 container with the password set. The -d flag runs it in the background and returns control to your terminal. List the running containers.

```
sania270804@ubuntu:~$ docker run -d -e POSTGRES_PASSWORD=password postgres:10.10
Unable to find image 'postgres:10.10' locally
10.10: Pulling from library/postgres
3106d02490d4: Pull complete
c5a7f905c8ec: Pull complete
16f2724dc303: Pull complete
ffc097878b09: Pull complete
756351b7a443: Pull complete
69802daaa561: Pull complete
88d1fc513b8f: Pull complete
f19250dffc5e: Pull complete
5c20c5b8227d: Pull complete
31babd3be108: Pull complete
0d24b08575ba: Pull complete
5a3f55930dd8: Pull complete
80369df48736: Pull complete
b18dd0a6efec: Pull complete
Digest: sha256:3263c8dad1b5aa6be80144786b18515910165ed771863d686c8b8fda30fea1ad
Status: Downloaded newer image for postgres:10.10
0cc379909b498df5a91f20111824ff37a6f826d5d7cfc57de8b665af7966019a
sania270804@ubuntu:~$ docker ps
CONTAINER ID   IMAGE                    COMMAND                  CREATED              STATUS
         PORTS
   NAMES
0cc379909b49   postgres:10.10           "docker-entrypoint.s…"   4 seconds ago        Up 2 second
s        5432/tcp
   cool_thompson
7e29138585bd   postgres:9.6             "docker-entrypoint.s…"   About a minute ago   Up About a
minute   5432/tcp
   strange_leakey
```

Start a PostgreSQL 10.10 container in the background. List the running containers again.

Task 3: Inspect Docker and Manage Containers and Images

```
sania270804@ubuntu:~$ docker info
Client:
 Version:    29.1.3
 Context:    default
 Debug Mode: false
 Plugins:
  buildx: Docker Buildx (Docker Inc.)
    Version:  0.30.1
    Path:     /usr/libexec/docker/cli-plugins/docker-buildx
  compose: Docker Compose (Docker Inc.)
    Version:  v5.6.1
    Path:     /usr/local/lib/docker/cli-plugins/docker-compose
  trust: Manage trust on Docker images (Docker Inc.)
    Version:  29.1.3
    Path:     /usr/libexec/docker/cli-plugins/docker-trust

Server:
 Containers: 5
  Running: 4
  Paused: 0
  Stopped: 1
 Images: 5
 Server Version: 29.1.3
 Storage Driver: overlayfs
  driver-type: io.containerd.snapshotter.v1
 Logging Driver: json-file
 Cgroup Driver: systemd
 Cgroup Version: 2
 Plugins:
  Volume: local
```

Display information about the Docker installation.

```
sania270804@ubuntu:~$ docker search ubuntu
NAME                      DESCRIPTION                                     STARS     OFFICIAL
ubuntu                    Ubuntu is a Debian-based Linux operating sys…   17878     [OK]
ubuntu/squid              Squid is a caching proxy for the Web. Long-t…   132
ubuntu/nginx              Nginx, a high-performance reverse proxy & we…   143
ubuntu/bind9              BIND 9 is a very flexible, full-featured DNS…   121
ubuntu/kafka              Apache Kafka, a distributed event streaming …   62
ubuntu/zookeeper          ZooKeeper maintains configuration informatio…   15
ubuntu/apache2            Apache, a secure & extensible open-source HT…   106
ubuntu/python             A chiselled Ubuntu rock with the Python runt…   34
ubuntu/prometheus         Prometheus is a systems and service monitori…   80
ubuntu/jre                Chiseled Java runtime based on Ubuntu. Long-…   25
ubuntu/dotnet-aspnet      Chiselled Ubuntu runtime image for ASP.NET a…   28
ubuntu/mysql              MySQL open source fast, stable, multi-thread…   74
ubuntu/dotnet-runtime     Chiselled Ubuntu runtime image for .NET apps…   24
ubuntu/postgres           PostgreSQL is an open source object-relation…   46
ubuntu/dotnet-deps        Chiselled Ubuntu for self-contained .NET & A…   16
ubuntu/redis              Redis, an open source key-value store. Long-…   26
ubuntu/jdk                Ubuntu rock for OpenJDK                         0
ubuntu/rabbitmq           Ubuntu-based RabbitMQ image - an open source…   0
ubuntu/memcached          Memcached, in-memory keyvalue store for smal…   8
ubuntu/grafana            Grafana, a feature rich metrics dashboard & …   13
ubuntu/loki               Grafana Loki, a log aggregation system like …   3
ubuntu/traefik            Ubuntu ROCK for Traefik, a modern HTTP rever…   3
ubuntu/mimir              Ubuntu ROCK for Mimir, a horizontally scalab…   0
ubuntu/php                PHP is a general-purpose scripting language …   10
ubuntu/grafana-agent      Ubuntu ROCK for Grafana Agent, an open-sourc…   0
sania270804@ubuntu:~$
```

Search Docker Hub for Ubuntu images.

```
sania270804@ubuntu:~$ docker images
                                                                  i Info →  U  In Use
IMAGE                                ID             DISK USAGE   CONTENT SIZE   EXTRA
intellij-idea-docker-intellij:latest f760d82771b1       11.6GB         3.01GB    U
lab5:1.0                             7f6559bd3775        417MB         85.1MB
postgres:10.10                       3263c8dad1b5        310MB         77.7MB    U
postgres:9.6                         caddd35b05cd        294MB         73.6MB    U
sonatype/nexus3:latest               91a9af44af77       1.81GB          809MB    U
sania270804@ubuntu:~$ docker run -d ubuntu
Unable to find image 'ubuntu:latest' locally
latest: Pulling from library/ubuntu
517b871c26ea: Pull complete
96aae9022e0e: Pull complete
e211d985152c: Download complete
Digest: sha256:065aa8626841f16057b78b027b464cd59c4fac36c204c72dcbf48d66f5fa8a96
Status: Downloaded newer image for ubuntu:latest
6b2c0b74475d03e47b9f1db1fbe693e1427ec8a0ff33953b44473cce9eadab24
sania270804@ubuntu:~$ docker ps
CONTAINER ID   IMAGE                    COMMAND                  CREATED         STATUS         PORTS
                                                                                 NAMES
0cc379909b49   postgres:10.10           "docker-entrypoint.s…"   4 minutes ago   Up 4 minutes   5432/tcp
                                                                                 cool_thompson
7e29138585bd   postgres:9.6             "docker-entrypoint.s…"   5 minutes ago   Up 5 minutes   5432/tcp
                                                                                 strange_leakey
651a396673d5   0b3563949d36             "/init"                  5 days ago      Up 7 minutes   3000/tcp, 0.0.
0.0:3001->3001/tcp, [::]:3001->3001/tcp                                          intellij-idea-docker-intellij-1
b3bcf179ef56   sonatype/nexus3:latest   "/opt/sonatype/nexus…"   5 days ago      Up 52 seconds  0.0.0.0:8007->
8007/tcp, [::]:8007->8007/tcp, 0.0.0.0:8006->8081/tcp, [::]:8006->8081/tcp   nexus
sania270804@ubuntu:~$ docker ps -a
CONTAINER ID   IMAGE                    COMMAND                  CREATED          STATUS                      P
ORTS                                                                                NAMES
6b2c0b74475d   ubuntu                   "/bin/bash"              12 seconds ago   Exited (0) 10 seconds ago
                                                                                    festive_hamilton
0cc379909b49   postgres:10.10           "docker-entrypoint.s…"   4 minutes ago    Up 4 minutes                5
432/tcp                                                                             cool_thompson
7e29138585bd   postgres:9.6             "docker-entrypoint.s…"   6 minutes ago    Up 6 minutes                5
432/tcp                                                                             strange_leakey
```

List local images, start an Ubuntu container, and compare the two container lists. Give the new Ubuntu container a moment to exit before running docker ps.

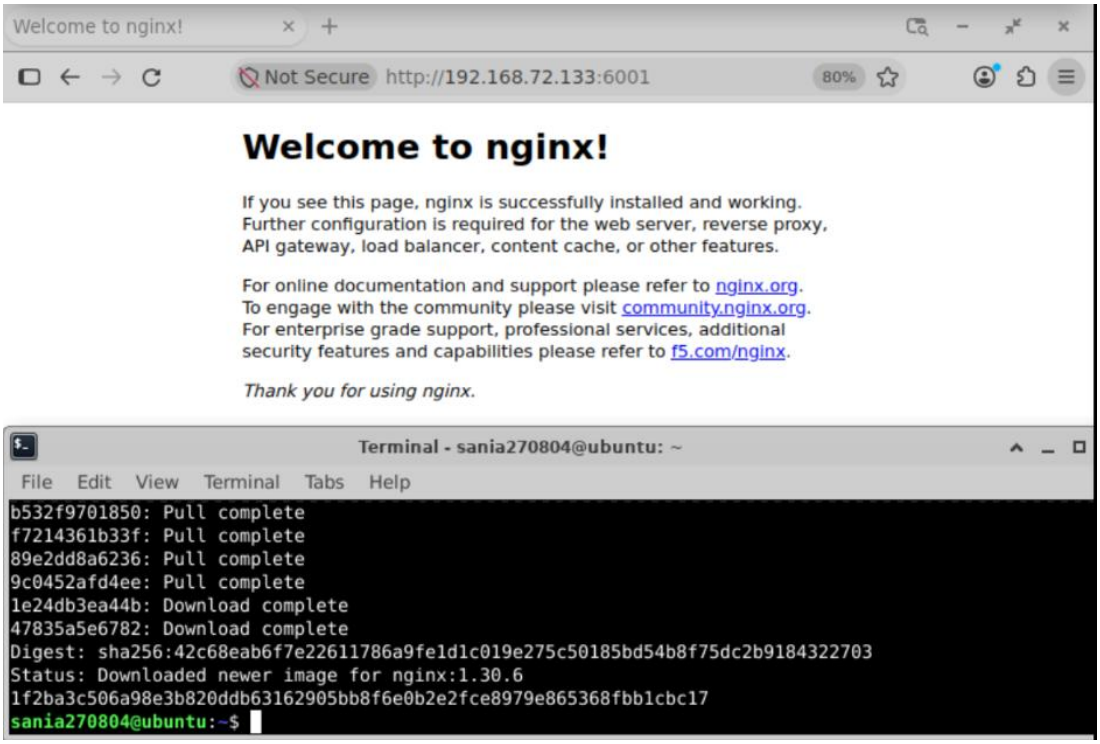
```
sania270804@ubuntu:~$ docker restart strange_leakey
strange_leakey
sania270804@ubuntu:~$ docker stop strange_leakey
strange_leakey
sania270804@ubuntu:~$ docker rm strange_leakey
strange_leakey
sania270804@ubuntu:~$ docker rmi postgres:9.6
Error response from daemon: conflict: unable to delete postgres:9.6 (must be forced) - container 53e165c86874 i
s using its referenced image caddd35b05cd
sania270804@ubuntu:~$ docker rmi -f postgres:9.6^C
sania270804@ubuntu:~$ docker rmi postgres:9.6
Error response from daemon: conflict: unable to delete postgres:9.6 (must be forced) - container 53e165c86874 i
s using its referenced image caddd35b05cd
sania270804@ubuntu:~$ docker rmi -f postgres:9.6
Untagged: postgres:9.6
```

Restart, stop and remove the PostgreSQL 9.6 container, then remove its image tag.

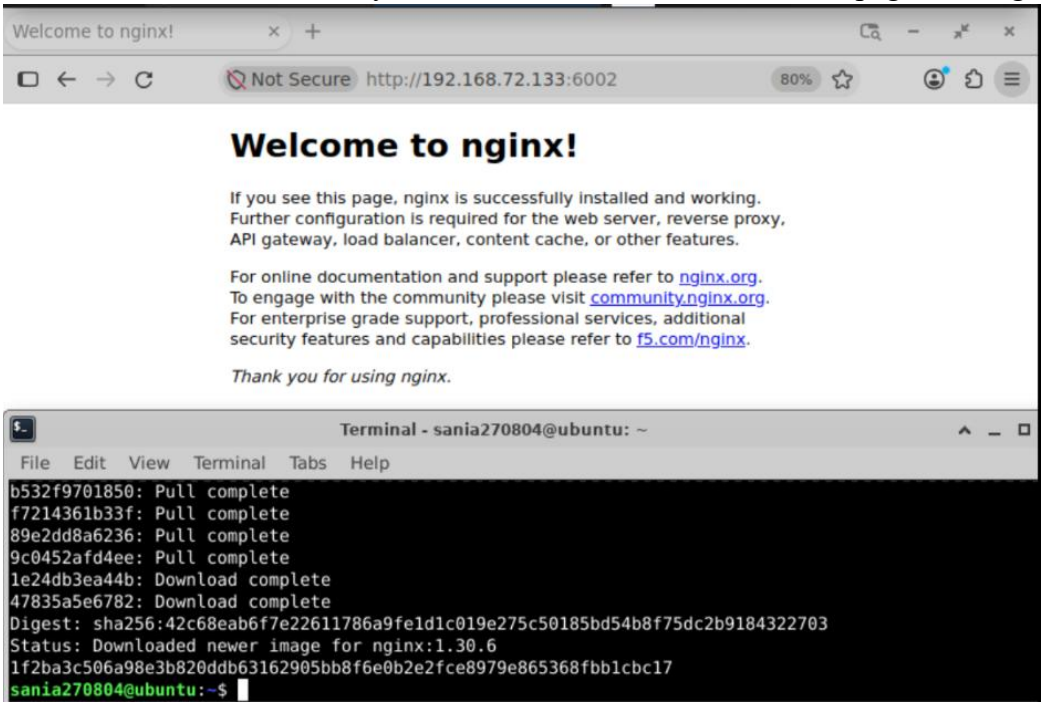
Task 4: Run Two NGINX Containers on Different Ports

```
sania270804@ubuntu:~$ docker run -dit -p 6001:80 --name nginx-latest nginx
Unable to find image 'nginx:latest' locally
latest: Pulling from library/nginx
1724c05f72c6: Pull complete
1962e1ff75b4: Pull complete
bbb080c71d50: Pull complete
37fb86c2a767: Pull complete
7cb4701ce7cd: Pull complete
6d2e43921e9e: Pull complete
512ac5e7ce8e: Pull complete
4582dace95e2: Download complete
d2c7e31f04a3: Download complete
Digest: sha256:165ddde8751e75a1c907d24ef10d9f6deb08b57e0fe56caf67512486aff76d82
Status: Downloaded newer image for nginx:latest
8ed1a741757d334502e0369965fbb41e65a45a9fc92223e1d1072bdcfebac4e3
sania270804@ubuntu:~$ docker run -d -p 6002:80 --name nginx-older nginx:1.30.6
Unable to find image 'nginx:1.30.6' locally
1.30.6: Pulling from library/nginx
2fc2bc44a8d3: Pull complete
cbbdc282ecf5: Pull complete
b532f9701850: Pull complete
f7214361b33f: Pull complete
89e2dd8a6236: Pull complete
9c0452afd4ee: Pull complete
1e24db3ea44b: Download complete
47835a5e6782: Download complete
Digest: sha256:42c68eab6f7e22611786a9fe1d1c019e275c50185bd54b8f75dc2b9184322703
Status: Downloaded newer image for nginx:1.30.6
1f2ba3c506a98e3b820ddb63162905bb8f6e0b2e2fce8979e865368fbb1cbc17
sania270804@ubuntu:~$
```

Each NGINX container listens on port 80 inside Docker. Publish them on different host ports so both pages can be opened from your browser.



Open http://<server-ip>:6001 in your browser. Replace <server-ip> with the address of your Ubuntu server. Confirm that you see the default NGINX welcome page from nginx-latest.



Open http://<server-ip>:6002 in another tab. Confirm that you see the default NGINX welcome page from nginx-older.

```
sania270804@ubuntu:~$ docker logs nginx-latest
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2026/11/14 10:33:05 [notice] 1#1: using the "epoll" event method
2026/11/14 10:33:05 [notice] 1#1: nginx/1.31.8
2026/11/14 10:33:05 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2026/11/14 10:33:05 [notice] 1#1: OS: Linux 7.0.0-43-generic
2026/11/14 10:33:05 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1024:524288
2026/11/14 10:33:05 [notice] 1#1: start worker processes
2026/11/14 10:33:05 [notice] 1#1: start worker process 29
2026/11/14 10:33:05 [notice] 1#1: start worker process 30
2026/11/14 10:33:05 [notice] 1#1: start worker process 31
2026/11/14 10:33:05 [notice] 1#1: start worker process 32
2026/11/14 10:33:05 [notice] 1#1: start worker process 33
2026/11/14 10:33:05 [notice] 1#1: start worker process 34
2026/11/14 10:33:05 [notice] 1#1: start worker process 35
2026/11/14 10:33:05 [notice] 1#1: start worker process 36
192.168.72.133 - - [14/Nov/2026:10:35:46 +0000] "GET / HTTP/1.1" 200 896 "-" "Mozilla/5.0 (X11; Ubuntu; Linux x
86_64; rv:159.0) Gecko/20100101 Firefox/159.0" "-"
2026/11/14 10:35:47 [error] 30#30: *1 open() "/usr/share/nginx/html/favicon.ico" failed (2: No such file or dir
ectory), client: 192.168.72.133, server: localhost, request: "GET /favicon.ico HTTP/1.1", host: "192.168.72.133
:6001", referrer: "http://192.168.72.133:6001/"
192.168.72.133 - - [14/Nov/2026:10:35:47 +0000] "GET /favicon.ico HTTP/1.1" 404 153 "http://192.168.72.133:6001
/" "Mozilla/5.0 (X11; Ubuntu; Linux x86_64; rv:159.0) Gecko/20100101 Firefox/159.0" "-"
sania270804@ubuntu:~$
```

View the logs from nginx-latest.

Task 5: Work Inside a Container and Save a New Image

```
sania270804@ubuntu:~$ docker attach nginx-latest
2026/11/14 10:52:52 [notice] 28#28: signal 28 (SIGWINCH) received
2026/11/14 10:52:52 [notice] 25#25: signal 28 (SIGWINCH) received
2026/11/14 10:52:52 [notice] 1#1: signal 28 (SIGWINCH) received
2026/11/14 10:52:52 [notice] 22#22: signal 28 (SIGWINCH) received
2026/11/14 10:52:52 [notice] 26#26: signal 28 (SIGWINCH) received
2026/11/14 10:52:52 [notice] 23#23: signal 28 (SIGWINCH) received
2026/11/14 10:52:52 [notice] 29#29: signal 28 (SIGWINCH) received
2026/11/14 10:52:52 [notice] 24#24: signal 28 (SIGWINCH) received
2026/11/14 10:52:52 [notice] 27#27: signal 28 (SIGWINCH) received
2026/11/14 10:52:52 [notice] 27#27: signal 28 (SIGWINCH) received
2026/11/14 10:52:52 [notice] 26#26: signal 28 (SIGWINCH) received
2026/11/14 10:52:52 [notice] 24#24: signal 28 (SIGWINCH) received
2026/11/14 10:52:52 [notice] 25#25: signal 28 (SIGWINCH) received
2026/11/14 10:52:52 [notice] 1#1: signal 28 (SIGWINCH) received
2026/11/14 10:52:52 [notice] 23#23: signal 28 (SIGWINCH) received
2026/11/14 10:52:52 [notice] 28#28: signal 28 (SIGWINCH) received
2026/11/14 10:52:52 [notice] 29#29: signal 28 (SIGWINCH) received
2026/11/14 10:52:52 [notice] 22#22: signal 28 (SIGWINCH) received
read escape sequence
sania270804@ubuntu:~$
```

Attach to the running NGINX process.

```
sania270804@ubuntu:~$ docker exec -it nginx-latest /bin/sh
# ls /
touch /usr/share/nginx/html/newfile.html
touch /usr/share/nginx/html/oldfile.html
rm /usr/share/nginx/html/oldfile.html
printenv
exitbin   dev             docker-entrypoint.sh  home  lib64  mnt  proc  run   srv  tmp  var
boot  docker-entrypoint.d  etc                   lib   media  opt  root  sbin  sys  usr
# # # # HOSTNAME=8ed1a741757d
HOME=/root
PKG_RELEASE=1~trixie
DYNPKG_RELEASE=1~trixie
ACME_VERSION=0.4.1
TERM=xterm
NGINX_VERSION=1.31.8
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin
NJS_VERSION=1.0.2
NJS_RELEASE=1~trixie
PWD=/
#
sania270804@ubuntu:~$
```

Open a separate shell inside the same container. List the root directory, create two files in the NGINX web directory, remove oldfile.html, and view the environment variables. exit closes this shell and returns you to Ubuntu.

```
sania270804@ubuntu:~$ docker diff nginx-latest
C /usr
C /usr/share
C /usr/share/nginx
C /usr/share/nginx/html
A /usr/share/nginx/html/newfile.html
C /run
A /run/nginx.pid
C /etc
C /etc/nginx
C /etc/nginx/conf.d
C /etc/nginx/conf.d/default.conf
C /var
C /var/cache
C /var/cache/nginx
A /var/cache/nginx/uwsgi_temp
A /var/cache/nginx/client_temp
A /var/cache/nginx/fastcgi_temp
A /var/cache/nginx/proxy_temp
A /var/cache/nginx/scgi_temp
sania270804@ubuntu:~$
```

On the Ubuntu server, compare the container filesystem with its starting state.

```
sania270804@ubuntu:~$ docker inspect nginx-latest
[
    {
        "Id": "8ed1a741757d334502e0369965fbb41e65a45a9fc92223e1d1072bdcfebac4e3",
        "Created": "2026-11-14T10:33:04.2053641592",
        "Path": "/docker-entrypoint.sh",
        "Args": [
            "nginx",
            "-g",
            "daemon off;"
        ],
        "State": {
            "Status": "running",
            "Running": true,
            "Paused": false,
            "Restarting": false,
            "OOMKilled": false,
            "Dead": false,
            "Pid": 13832,
            "ExitCode": 0,
            "Error": "",
            "StartedAt": "2026-11-14T10:52:49.977884687Z",
            "FinishedAt": "2026-11-14T10:49:56.716501193Z"
        },
        "Image": "sha256:165ddde8751e75a1c907d24ef10d9f6deb08b57e0fe56caf67512486aff76d82",
        "ResolvConfPath": "/var/lib/docker/containers/8ed1a741757d334502e0369965fbb41e65a45a9fc92223e1d1072bdcfebac4e3/resolv.conf
",
        "HostnamePath": "/var/lib/docker/containers/8ed1a741757d334502e0369965fbb41e65a45a9fc92223e1d1072bdcfebac4e3/hostname",
        "HostsPath": "/var/lib/docker/containers/8ed1a741757d334502e0369965fbb41e65a45a9fc92223e1d1072bdcfebac4e3/hosts",
        "LogPath": "/var/lib/docker/containers/8ed1a741757d334502e0369965fbb41e65a45a9fc92223e1d1072bdcfebac4e3/8ed1a741757d334502
e0369965fbb41e65a45a9fc92223e1d1072bdcfebac4e3-json.log",
        "Name": "/nginx-latest",
        "RestartCount": 0,
        "Driver": "overlayfs",
        "Platform": "linux",
        "MountLabel": "",
        "ProcessLabel": "",
```

Inspect the container's configuration and state.

```
sania270804@ubuntu:~$ docker commit nginx-latest nginx:myimage
sha256:77a2feaacc26fd19b6b301c9c9b047704cd5417626d162d50ee4178035dd6bf9
sania270804@ubuntu:~$
```

Save the container's filesystem changes as a new local image.